EdTech Data Analysis using Azure SQL & Azure Data Factory

Problem Statement

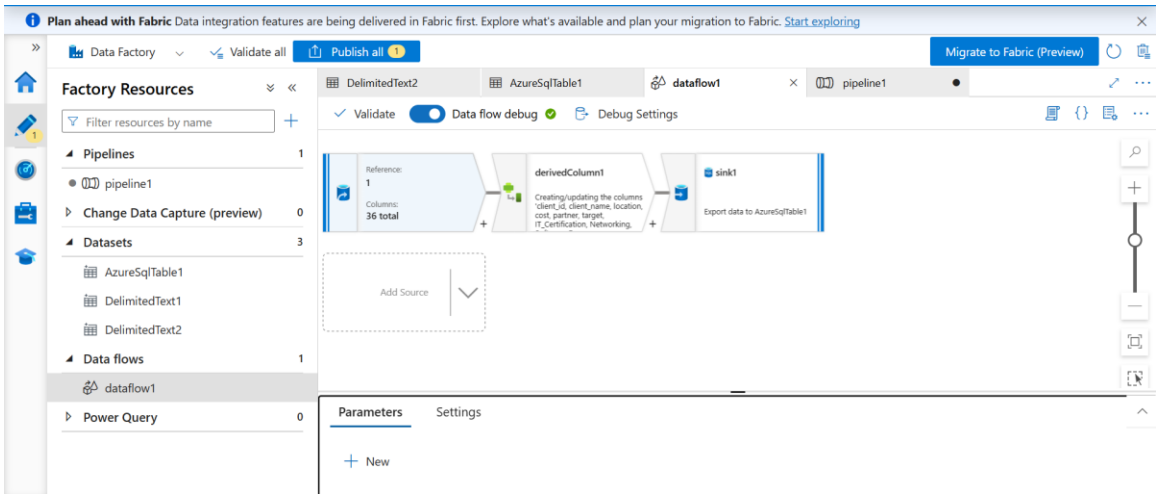
SkillBridge Learning aims to analyze client, course, service, and workforce data to align training programs with industry requirements. The challenge is to identify suitable candidates, optimize services, and improve deployment strategies using structured data.

Objective

The objective of this project is to utilize Azure SQL Database and Azure Data Factory to store, transform, and analyze client data. The goal is to derive insights for client classification, course demand, regional strategy, cost optimization, and workforce deployment.

Data Ingestion and Transformation

In this step, the dataset is ingested and transformed using Azure Data Factory. The source data is read from a delimited file containing multiple columns, where all values are initially in string format. A Mapping Data Flow is implemented to perform data transformation, where a Derived Column transformation is used to convert string values into appropriate integer data types using functions like toInteger() and to handle null values using conditional expressions. After transformation, the cleaned and structured data is loaded into Azure SQL Database. This step ensures that the data is consistent, error-free, and ready for further analysis and use case implementation.



Use Case 1: Client Classification

Objective: Classify clients based on partner type, cost, and domain-specific services.

Explanation: This helps identify high-value enterprise clients, flexible SME/startup clients, and domain-specific clients such as retail and telecom.

Code:-

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explore

SQL query 1

Run | Save as view

```sql
SELECT
    client_name,
    partner,
    cost,
    CASE
        WHEN partner LIKE '%Enterprise%' AND cost >= 15000 THEN 'Enterprise High Value'
```

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explore

SQL query 1

Run | Save as view

```sql
        WHEN partner LIKE '%SME%' OR partner LIKE '%Startup%' THEN 'Flexible Clients'
        WHEN Retail_POS_Systems = 1 THEN 'Retail Client'
        WHEN Telecom_Solutions = 1 THEN 'Telecom Client'
        ELSE 'General Client'
    END AS category
FROM All_Data;
```

Output:-

Messages | Results

	client_name	partner	cost	category
1	IPMC Ghana	Corporate	10000	General Client
2	Deans Computer Services	SMEs	20000	Flexible Clients
3	Mitsubishi Electric Information Net...	Enterprises	20000	Enterprise High Value
4	Michell Consulting Group	Enterprises	20000	Enterprise High Value

Messages | Results

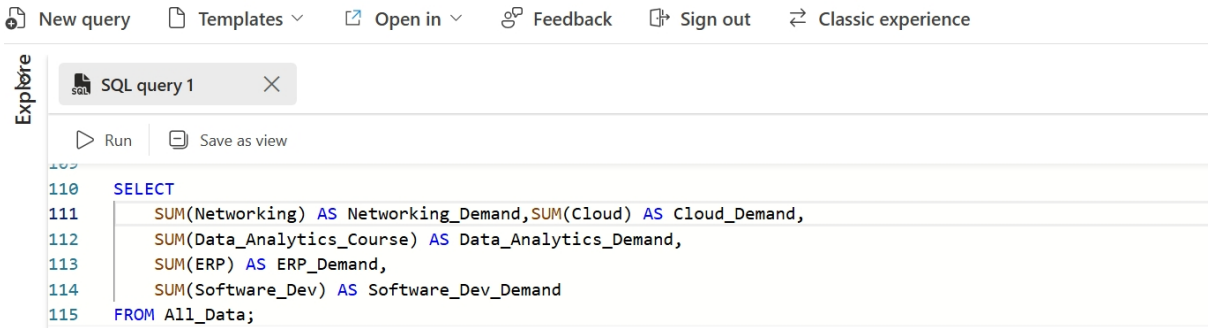
	client_name	partner	cost	category
7	Inconnect	Telecom Companies	10000	Telecom Client
8	Take2	SMEs	4000	Flexible Clients
9	Dogwood Management Partners	Enterprises	20000	Enterprise High Value
10	Rapid POS	Retail Businesses	10000	Retail Client

Use Case 2: Course Demand Analysis

Objective: Analyze which courses and skills are most demanded by clients.

Explanation: This helps prioritize training programs such as Networking, Cloud, Data Analytics, and ERP based on industry demand.

Code:-



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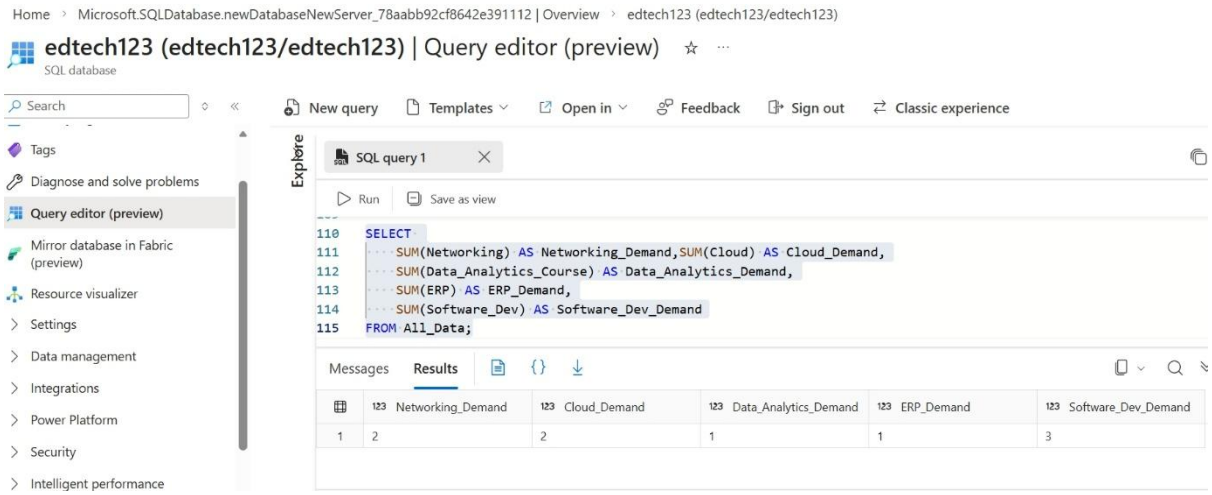

Use Case 3: Regional Strategy Planning

Objective: Analyze client distribution and cost patterns across locations.

Explanation: This helps identify high-value regions, cost-sensitive markets, and enables region-specific strategies.

Code:-

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explore

SQL query 1

Run | Save as view

```sql
SELECT
    location,COUNT(*) AS total_clients,AVG(cost) AS avg_cost,
    CASE
        WHEN AVG(cost) >= 15000 THEN 'High Value Region'
        WHEN AVG(cost) BETWEEN 5000 AND 15000 THEN 'Medium Market'
        ELSE 'Cost Sensitive Region'
```

New query | Templates | Open in | Feedback | Sign out | Classic experience

Explore

SQL query 1

Run | Save as view

```sql
        WHEN AVG(cost) >= 15000 THEN 'High Value Region'
        WHEN AVG(cost) BETWEEN 5000 AND 15000 THEN 'Medium Market'
        ELSE 'Cost Sensitive Region'
    END AS region_category
FROM All_Data
GROUP BY location;
```

Output:-

Messages | Results

	location	total_clients	avg_cost	region_category
1	Europe	1	10000	Medium Market
2	Ghana	1	10000	Medium Market
3	India	1	4000	Cost Sensitive Region
4	Japan	1	20000	High Value Region

Messages | Results

	location	total_clients	avg_cost	region_category
5	Middle East	1	10000	Medium Market
6	South Africa	1	4000	Cost Sensitive Region
7	UK	1	20000	High Value Region
8	USA	3	16666	High Value Region

Use Case 4: Cost vs Service Optimization

Objective: Map client cost categories with the type of services utilized.

Explanation: This helps optimize service offerings such as training, IT services, and consulting based on budget levels.

Code:-

```sql
SELECT
    client_name,cost,CASE
        WHEN cost < 5000 AND Training = 1 THEN 'Training Focus'
        WHEN cost BETWEEN 5000 AND 15000 AND IT_Services = 1 THEN 'IT Services Focus'
        WHEN cost >= 15000 AND Consulting_Service = 1 THEN 'Consulting Focus'ELSE 'Mixed Services'
    END AS service_strategy
FROM All_Data;
```

Output:-

	client_name	cost	service_strategy
1	IPMC Ghana	10000	Mixed Services
2	Deans Computer Services	20000	Mixed Services
3	Mitsubishi Electric Information Network	20000	Mixed Services
4	Michell Consulting Group	20000	Consulting Focus

	client_name	cost	service_strategy
7	Inconnect	10000	Mixed Services
8	Take2	4000	Training Focus
9	Dogwood Management Partners	20000	Mixed Services
10	Rapid POS	10000	IT Services Focus

Use Case 5: Workforce Targeting & Deployment

Objective: Align workforce categories with deployment strategies.

Explanation: This ensures proper placement of freshers, engineers, analysts, and retail staff into suitable roles.

Code:-

```sql
SELECT
    client_name,target,CASE
        WHEN target LIKE '%Freshers%' THEN 'Training & Placement'
        WHEN target LIKE '%Engineers%' OR target LIKE '%Analysts%' THEN 'Consulting & Analytics'
        WHEN target LIKE '%Retail%' THEN 'Retail POS Deployment'
        ELSE 'General Deployment'
```

```sql
        WHEN target LIKE '%Freshers%' THEN 'Training & Placement'
        WHEN target LIKE '%Engineers%' OR target LIKE '%Analysts%' THEN 'Consulting & Analytics'
        WHEN target LIKE '%Retail%' THEN 'Retail POS Deployment'
        ELSE 'General Deployment'
    END AS deployment_strategy
FROM All_Data;
```

Output:-

	client_name	target	deployment_strategy
1	IPMC Ghana	Freshers	Training & Placement
2	Deans Computer Services	Junior Developers	General Deployment
3	Mitsubishi Electric Information Network	Engineers	Consulting & Analytics
4	Michell Consulting Group	Analysts	Consulting & Analytics

	client_name	target	deployment_strategy
7	Inconnect	Network Engineers	Consulting & Analytics
8	Take2	Freshers	Training & Placement
9	Dogwood Management Partners	Managers	General Deployment
10	Rapid POS	Retail Tech Staff	Retail POS Deployment

Conclusion

This project demonstrates how Azure SQL Database and Azure Data Factory can be used to transform raw data into meaningful business insights. By analyzing client, course, service, and workforce data, organizations can improve decision-making, optimize training programs, and enhance placement success.